

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- the predicted value of y for $x_0 = 4$.
- the $se(f)$ corresponding to part (a).
- a 95% prediction interval for y given $x_0 = 4$.
- a 99% prediction interval for y given $x_0 = 4$.
- a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

a. The fitted least squares line : $\hat{y}_i = 1.2 + 0.8x_i$

$$x_0 = 4 \Rightarrow \hat{y}_0 = 4.4$$

b.

$$se(f) = s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

$$\bar{x} = 1$$

$$s = \sqrt{\frac{SSE}{n-2}}$$

$$s = \sqrt{1.2}$$

$$\Rightarrow se(f) = \sqrt{1.2} \cdot \sqrt{\frac{1}{5} + \frac{(4-1)^2}{10}} \approx 1.1489$$

c.

$$S_{pred} = s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

$$\Rightarrow S_{pred} = \sqrt{1.2} \cdot \sqrt{1 + \frac{1}{5} + \frac{(4-1)^2}{10}} \approx 1.5875$$

95% prediction interval (P.I.) for $\hat{y}_0$.

$$\hat{y}_0 \pm t_{0.025, 5-2} \cdot S_{pred}$$

$$= [-1.052, 9.052]$$

d. 99% P.I. for $\hat{y}_0$:

$$\hat{y}_0 \pm t_{0.005, 5-2} \cdot S_{pred}$$

$$= [-5.272, 13.272]$$

e. $\bar{x} = 1$

$$\Rightarrow S_{\text{pred}} = \sqrt{1.2} \cdot \sqrt{1 + \frac{1}{5} + \frac{(1-1)^2}{10}} = 1.2$$

95% P.I. for y given $x = \bar{x}$:

$$\hat{y} \pm t_{0.025, 5-2} \cdot S_{\text{pred}} \\ = [-2.819, 4.819]$$

The width of the interval in e. is smaller than the interval in c.

This because when $x_0 = \bar{x}$, it is closet to the sample center and the uncertainty is minimized.